

# ABRAHAM OGOL

**AI & DATA TRANSFORMATION LEAD | DATA ENGINEERING | AUTOMATION | AWS | AI AGENTS**

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## **PROFESSIONAL SUMMARY**

AI, data and technology transformation leader with 11+ years of experience combining software engineering, enterprise architecture, data platforms, automation, cloud, analytics and business transformation. Strong background translating operational pain points into scalable digital solutions, with hands-on depth across Python, SQL, APIs, event-driven systems, AWS, databases and AI/LLM-enabled workflows. Experienced leading cross-functional initiatives from current-state analysis and solution design through implementation, governance, adoption and continuous improvement.

Experienced across Python, SQL, PostgreSQL, MySQL, SQL Server, MongoDB, Apache Kafka, message brokers, asynchronous processing, REST APIs, microservices, event-driven architecture, AWS, Azure, GCP, Docker, Kubernetes, CI/CD, data integration, schema design, operational analytics, IAM, role-based access controls, observability and secure software delivery. Comfortable balancing performance, reliability, scalability, data quality, confidentiality, compliance and cost.

Business- and data-minded technology leader who works closely with operations, finance, product, security, customer-facing teams and executives to understand how work actually happens before automating it. Experienced mapping processes, identifying failure points and manual controls, defining future-state workflows, evaluating AI use cases, building data-driven platforms, mentoring teams and communicating complex technical choices in clear business terms.

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## **CORE AI, DATA TRANSFORMATION & TECHNICAL COMPETENCIES**

### **AI & Data Transformation Strategy**

**AI Roadmaps • Use-Case Identification • Feasibility Analysis • Business Process Analysis • Current-State Mapping • Future-State Design • Workflow Automation • Transformation Planning • Pilot-to-Scale Delivery • Benefits Tracking**

### **Data Engineering, Streaming & APIs**

**Python • SQL • Apache Kafka • Event-Driven Architecture • Message Brokers • Asynchronous Processing • REST APIs • OpenAPI Concepts • Microservices • Batch & Near-Real-Time Pipelines • Data Integration • API Integration • Validation • Failure Handling**

### **AI, Automation & Intelligent Workflows**

**Generative AI • LLM Applications • AI Agents • Agentic Workflows • RAG • MCP-Style Tool Integration • AI-Assisted Development • Prompt/Context Engineering • Human-in-the-Loop Controls • Anomaly Detection Concepts • Responsible AI**

### **Cloud Data Platforms & Infrastructure**

**AWS • Azure • GCP • Cloud-Native Architecture • Data Lakes • Analytics Layers • Docker • Kubernetes • Infrastructure as Code • CI/CD • Linux • Production Systems • Deployment Automation • Performance & Cost Optimisation**

### **Data Quality, Monitoring & Reliability**

**Data Quality Checks • Validation Rules • Reconciliation • Schema Evolution • Monitoring •**

## **SELECTED AI, DATA & TRANSFORMATION IMPACT**

- Scaled an engineering and technology organization from 5 to 50+ professionals, strengthening hiring, onboarding, engineering standards, and delivery structures while reducing onboarding time by 40%.
- Built and operated API, database, microservice and cloud platforms in cross-functional teams spanning engineering, QA, product, DevOps, operations and customer-facing functions.
- Improved software platform performance by 50% while reducing infrastructure costs

Logging • Observability • Root-Cause Analysis • Anomaly Detection • Backup & Recovery • Disaster Recovery • Business Continuity • Performance Optimisation

#### Data Modelling & Governance

PostgreSQL • SQL Server • MySQL • MongoDB • Database Design • Data Modelling • Master Data Concepts • Data Dictionaries • Schema Management • Query Optimisation • Transactions • Data Lineage • Data Flows • Data Governance

#### Security, Privacy & Compliance

IAM • Role-Based Access Control • Authentication • Authorization • Encryption Concepts • Secure APIs • DevSecOps • Data Confidentiality • Kenya Data Protection Act Awareness • AI Risk • Explainability • Bias & Drift Risk • Regulatory Compliance • Audit Readiness

#### Business Analysis & Process Automation

Requirements Discovery • Process Mapping • Pain-Point Analysis • Future-State Workflows • SOP Improvement • Workflow Automation • Operational Analytics • Business Cases • Vendor Evaluation • Change Enablement • Continuous Improvement

#### Leadership, Governance & Stakeholder Management

Programme Leadership • Senior Stakeholder Management • C-Level Communication • Mentoring • Technical Direction • Risk Management • Multi-Country Delivery • Vendor & Partner Engagement • Cross-Functional Leadership • Agile • Governance • Decision Support

#### Insurance & Operational Use-Case Alignment

Claims Analytics Concepts • Fraud Detection Concepts • Regulatory Reporting • Actuarial Data Support • Operational Dashboards • Member/Policy/Claims/Provider/Payment Data Models • Financial Integrations • Healthcare & Sensitive-Data Workflows

by 20% through architecture, application, and infrastructure optimization.

- Established CI/CD, deployment automation, and engineering practices that reduced deployment cycles to under 10 minutes and enabled more frequent, reliable releases.
- Introduced automated testing and engineering quality practices that contributed to a 45% reduction in production defects.
- Led the design and delivery of technology platforms supporting 250+ customers across five African markets and integrating 6,000+ network nodes/devices.
- Designed, built, tested, deployed and monitored scalable APIs, database-driven applications, distributed services, automation workflows and cloud platforms in production.

## PROFESSIONAL EXPERIENCE

### CHIEF TECHNOLOGY OFFICER

Cognativ Technology Ltd | March 2025 – Present

Lead technology, AI, data and automation initiatives across enterprise platforms, APIs, databases, cloud environments, engineering teams and operational workflows.

#### Key Responsibilities & Contributions

- Lead discovery and technical design for data, automation and AI-enabled initiatives, translating operational requirements into scalable architectures, integrations, implementation plans and measurable outcomes.
- Mentor engineers and technical teams across Python, APIs, databases, data integration, automation, debugging, architecture and production engineering while strengthening ownership and delivery quality.
- Define standards covering data quality, API design, testing, CI/CD, observability, security, documentation, access control, production readiness and operational continuity.
- Guide architecture decisions for data platforms, APIs, distributed systems, event-driven services, cloud workloads and database-backed applications, balancing scale, reliability, security, maintainability, data quality and cost.
- Lead architecture and quality reviews focused on data flows, integration patterns, schema design, scalability, security, resilience, monitoring and production reliability.
- Evaluate and integrate AI/LLM capabilities, intelligent automation and emerging technologies based on business value, technical fit, data readiness, reliability, explainability, security, scalability and responsible-use considerations.

## EDUCATION

Bachelor of Science in  
Information Technology  
Maseno University

## CERTIFICATIONS

- Certified Kubernetes Administrator (CKA)
- Docker Certified Associate (DCA)
- Microsoft Azure Architect Design
- AWS Certified Cloud Practitioner
- Cisco Certified Design Professional
- Zscaler Internet Access (ZIA) Certifications

- Partner with business, product, engineering, operations and customer-facing stakeholders to map problems, assess options, prioritise transformation opportunities and translate them into delivery roadmaps and production-ready solutions.
- Remain technically hands-on in Python, SQL, API/data integration, database design, automation, architecture reviews, troubleshooting, performance analysis and production support.
- Zscaler Private Access (ZPA) Certifications

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## PRODUCT DEVELOPMENT SPECIALIST

Go City | August 2024 – February 2025

Delivered product and data-enabled capabilities end to end across requirements analysis, process understanding, technical design, integration, testing, release readiness and post-launch improvement.

- Coordinated delivery across engineering, QA, product and business stakeholders, maintaining alignment between operational needs and technical execution.
- Led requirements analysis and translated business processes, user needs and operational constraints into implementable technology solutions.
- Supported architecture and technology decisions to ensure functional, operational, and scalability requirements were addressed.
- Improved engineering workflows, release practices, technical documentation, backlog execution, and delivery processes.
- Managed integration activities, data exchange between systems, feedback cycles, release readiness and handover to improve solution quality, adoption and customer outcomes.

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## PRODUCTS DEVELOPMENT LEAD

CSquared | May 2022 – May 2024

Led the development and evolution of data-driven digital platforms supporting enterprise services across multiple African markets.

### Key Responsibilities & Contributions

- Led API-driven distributed platforms deployed across five African markets, working across cloud architecture, integrations, databases, event-driven services, container environments, operational analytics and production operations.
- Worked with developers, product managers, QA, architects, infrastructure engineers and customer-facing teams to design, build, monitor and improve platform services using operational, performance and customer data.
- Scaled the technology organization from 5 to 50+ professionals, strengthening hiring, onboarding, engineering standards, capability development, and delivery accountability.
- Improved platform and data-service quality through architecture reviews, automated testing, delivery standards, monitoring, technical documentation and continuous improvement.
- Led the development of an innovative connectivity-on-demand platform leveraging **Software-Defined Networking**.
- Led development of a hardware-independent **SD-WAN platform** using universal CPE architecture.
- Supported scalable platforms serving 250+ enterprise customers across five African markets, integrating thousands of network nodes/devices and generating operational data used for service monitoring and decision-making.
- Improved platform performance by 50% and reduced infrastructure costs by 20% through architecture, resource, cloud and operational optimisation.

- Led geographically distributed technical teams and coordinated architecture, delivery, production support, and customer-facing priorities across markets.
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## **MANAGER – ENTERPRISE SOLUTIONS DESIGN & IMPLEMENTATION**

### **Safaricom PLC | March 2021 – July 2022**

Led enterprise solution design and implementation across cloud, data, APIs, systems integration, security, IoT and complex customer environments.

- Designed end-to-end enterprise solutions spanning AWS, Azure, VMware, OpenStack, APIs, data flows, connectivity, IoT, security and integrated business systems.
  - Consulted directly with enterprise and public-sector customers, sales teams and internal stakeholders to understand business processes, data, integration, security and deployment requirements and turn them into scalable technical solutions.
  - Led solution architecture and technical integration activities for complex enterprise environments, including API patterns, system topology, cloud/hybrid deployment choices, and implementation sequencing.
  - Led and mentored architects and technical specialists, providing architecture guidance, implementation support, technical reviews, escalation leadership, and problem resolution across customer environments.
  - Collaborated with direct sales, engineering, integration, networking, and customer teams from pre-sales scoping and solution presentations through implementation and technical handover.
  - Developed technical specifications, integration and data-flow designs, security controls, deployment considerations and implementation approaches for custom development and enterprise transformation requirements.
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## **IoT ENGINEER – PLATFORMS & PRODUCT DEVELOPMENT**

### **Safaricom PLC | July 2019 – April 2021**

Designed, developed, tested and supported API-driven, database-backed and cloud-connected platforms integrating operational data, automation and enterprise workflows.

- Designed and implemented a SIM lifecycle platform automating activation, suspension, billing and customer workflows, integrating operational and customer data across enterprise systems.
  - Designed and deployed IoT hardware and platform capabilities.
  - Developed a telecom base-station management platform using integrated operational data and workflow automation, contributing to a 25% reduction in operational costs.
  - Designed API-first platforms incorporating workflow automation, data orchestration, enterprise integration, database services and operational analytics.
  - Automated customer and operational workflows, contributing to a **12.5% revenue increase**.
  - Provided hands-on software development leadership, production troubleshooting, architecture guidance, code-quality support, and mentoring to engineers.
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## **PYTHON SOFTWARE LEARNING FACILITATOR & DEVELOPER**

### **Andela | March 2018 – June 2019**

- Developed, tested and supported Python applications, REST APIs, data-processing services, database solutions and internal engineering tools.

- Mentored and trained 15+ software developers monthly in Python, APIs, databases, software design, testing, debugging and analytical problem-solving.
  - Supported hands-on backend development, debugging, code reviews, API design, unit testing, integration testing, and software quality.
  - Applied Agile development practices and supported developer growth.
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## **PYTHON SOFTWARE DEVELOPER TEAM LEAD**

**AARL | 2015 - 2018**

- Led hands-on software and data development using Python, designing REST APIs, data models, databases and maintainable services from requirements through deployment.
- Designed application data models and databases.
- Led sprint planning, technical discussions and software delivery.
- Mentored engineers on architecture, debugging and technical problem-solving.